A Study on Distributed Model Predictive Control in Longitudinal Traffic with CACC Platooning Systems

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***Abstract*—With the recent advancements in automated driving technology, self-driving cars can interact with each other and form a platoon on a highway. The vehicle at the front (lead vehicle) of the platoon controls the speed of platoon vehicles, making them drive safely in close proximity on the highway. An onboard computer is equipped with a vehicle-to-vehicle (V2V) communication device for automated driving systems to form a platoon. This device can transmit and receive data from one car to another in a platoon using dedicated short-range communication. V2V technology allows cars to share information at a lower latency rate. This computer in the vehicle would not only control braking and acceleration but also use the front vehicle's camera/radar/lidar data to create a Cooperative Adaptive Cruise Control. We go through some control strategies applied to the application. Ultimately, we implement Distributed model-based predictive control (MPC) and Centralized MPC as our strategies based on a simple vehicle model for longitudinal motion to achieve simulation.**

***Index Terms*—Platooning, V2V technology, Model Predictive Control, Cooperative Adaptive Cruise Control**

I. Introduction

The vehicle platooning system uses V2X communication technology to group multiple vehicles to drive in a queue to improve traffic efficiency. Platoon technology has progressively advanced from cruise control (CC) technology to cooperative adaptive cruise control (CACC) technologies to achieve a higher level of autonomy specified by SAE standards. In CACC systems, the leader vehicle actively communicates with the following vehicles by sending its speed, accelerator, and brake signals to decide on longitudinal control reasonably. Longitudinal spacing control and communication technologies are the primary focus for platoon driving. Currently, platoon technology has developed to an extent of large scale commercial freight.

The control of platoon systems is categorized into two - centralized and distributed control. For platoon modelling of CACC control, many research works have used a predecessor-following (PF) communication topology [1]. The scenario we have chosen has multiple vehicles in the same lane forming one single platoon by adjusting inter-vehicle distances and using longitudinal control based on information from leading/neighbouring vehicles to maintain the desired speed. Fatemeh has shown that distributed algorithm scales well with an increasing number of vehicles in platoon [2]. The proposed algorithm has decomposed the single optimal problem into multiple local problems, associated with each vehicle. Yang has considered the vehicles in a platoon to be dynamically decoupled but constrained by spatial geometry. The vehicle is assigned a local optimal control problem based on neighbouring vehicles where the cost function penalizes the error between predicted and assumed trajectories [3]. The course project is an exploration to DMPC for longitudinal control within CACC platooning- the aim is to provide the trade-offs, benefits, and challenges associated with distributed control by comparing it with a centralized control. This will aid in optimizing the traffic flow, increasing the road safety leading to a more robust deployments of autonomous vehicles for the future transportation/logistics industry.

II. Literature Review: Different Control Strategies

The Control Design for platooning should focus on meeting the vehicle dynamics, spacing policy, information flow topology from the available communication system and stability requirements. Mostly, the control design considers a leader-follower approach where the leader vehicle determines the speed of the platoon, and each follower satisfies the spacing constraints with the predecessor. In a few research works [7], platoon control considers the lead vehicle to be exogenous to the platoon controller - either controlled by a separate ACC or a manual driver.

Various control design approaches are proposed in the research community, such as classic Linear Quadratic Regulator (LQR), MPC and Proportional Integral Derivative (PID). We primarily focus on MPC algorithms, which optimize a finite horizon cost function with hard constraints such as minimum safe inter-vehicle distance and vehicle speed. These constraints will aid in determining the desired control actions so that the vehicle is operated within its performance limit or to prevent accidents. The MPC formulation can be in a distributed approach where the lead vehicle provides an input reference to the platoon. This flexible approach allows a control design to split and actuate each follower vehicle based on their front vehicle. In a centralized control design, all the platoon information and platoon references are utilized to design the control actions collectively for all the vehicles in the platoon.

III. System Model

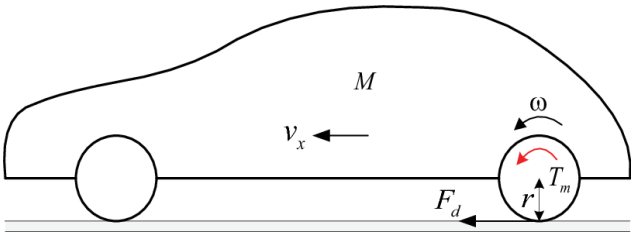


Fig. 1. Schematic representation of the vehicle model.

The vehicle is assumed to be a lumped parameter model, which simplifies the complexity of the system by not taking into account the structural details of the powertrain. It is represented as a two-dimensional rectangle with wheels, where the application of torque (T_m) to the wheels results in linear acceleration or deceleration. The forces acting on the vehicle include the driving force (F_d) and the resistance forces, which are functions of the vehicle's speed (v_x) and other environmental factors.

A. Longitudinal dynamics

Vehicle motion, considering the longitudinal dynamics, is determined by

$$\frac{ds_h}{dt} = V_h \tag{1}$$

$$\frac{dv_h}{dt} = a_h \tag{2}$$

where S_h, is the host vehicle's longitudinal position ,v_h is the longitudinal speed and a_h is the longitudinal acceleration. According to newton's second law, the longitudinal dynamics of the Vehicle are modeled as follows.

$$a_h = \frac{1}{m_{h_e}} \left(\frac{T_a}{r_w} - (R_r(V_h) + R_a(V_h) + R_g(S_h)) \right) \tag{3}$$

where m_h is the mass of the host vehicle, m_{h_e} is the effective mass which is generally considered to be 1.1 times the mass of the vehicle m_h. T_a is the total torque applied at the drive wheels, which includes both the driving force and braking force, and $R_r(V_h), R_a(V_h), R_g(S_h)$ are the rolling resistance, aerodynamic drag, and gradient resistance respectively.

TABLE I
Vehicle Parameters

Parameters	Values
Vehicle Mass, m_h	1538 kg
Wheel Radius, r_h	0.296 m
Air Density at Sea Level, ρ_{air}	1.225 kg/m^3
Frontal Area, A_h	2.2 m^2
Air Drag Coefficient, C_{d_h}	0.32
Rolling Resistance Coefficient, f_h	0.015
Acceleration due to Gravity, g	9.81 m/s^2

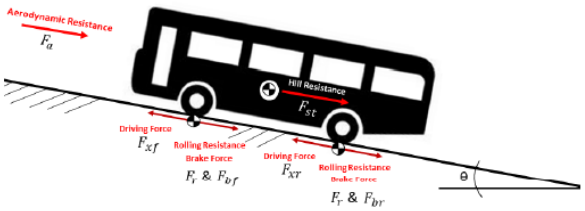


Fig. 2. Resistance forces on a Vehicle

B. Aerodynamic Resistance:

Aerodynamic Resistance is the force exerted on a body whenever there is a relative velocity between the body and the air and can be expressed using the formula

$$F_{aero} = \frac{1}{2}\rho C_d d_h A_h (V + V_0)^2 \tag{4}$$

where F_{aero} is the Aerodynamic force, ρ is the density of the air, C_d, Aerodynamic drag Coefficient, A_h, Frontal projection area of the Vehicle, V_0 is the wind speed opposite to the direction of vehicle movement.

C. Rolling Resistance:

Tire rolling Resistance is the energy that the Vehicle needs to overcome to maintain movement at a consistent speed over a surface and can be expressed using the formula.

$$F_{Roll} = m_h g cos\phi(C_r) \tag{5}$$

Where F_{Roll} is the Rolling resistance force, g Acceleration due to gravity, m_h, Mass of the host vehicle, ϕ is the road grade expressed in angle. Since we are considering the urban driving scenario, the road grade is considered as zero, $\phi = 0$, V_h is the host vehicle speed, $C_r = 0.015$ is the coefficient of rolling resistance.

D. Gradient Resistance:

Gradient resistance is a resistance force that occurs due to the movement of the Vehicle on an inclined surface and is opposite to the direction of movement of the Vehicle. This Resistance is directly proportional to the Mass of the Vehicle and the slope of the hill.

$$F_{Grade} = m_h g sin\phi \tag{6}$$

where, F_{Grade} is the Gradient resistance force, g Acceleration due to gravity,m_h, mass of the host vehicle,ϕ is the road grade expressed in angle. Since we are considering the urban driving scenario, the road grade is considered zero,

IV. High-Level Control Problem for Vehicle Platooning

In the platooning scenario depicted in Fig 3, our control problem is precisely defined in terms of state and action variables.

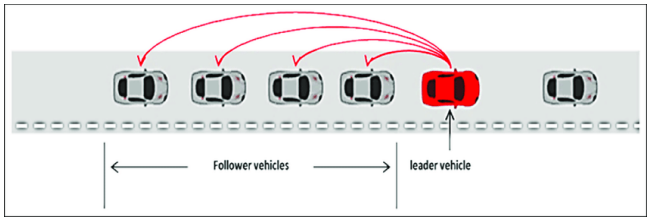


Fig. 3. High Level Structure of Platooning Vehicles

A. State Definition

The state of each vehicle within the platoon is defined by its velocity. It is assumed that the velocity and other necessary information for control are accurately and reliably obtained. Moreover, we incorporate a one-step delay in observation to emulate real-world sensing and processing constraints.

B. Action Definition

The control actions are:

- Braking Torque (T_b): The negative torque applied to the vehicle's wheels to reduce its speed.
- Traction Torque (T_a): The positive torque applied to increase or maintain the vehicle's speed.

However to implement the MPC, a discretization of the state equations is required. Thus, using Euler discretization, the discrete state equations were obtained as follows:

$$s(i+1) = s(i) + v(i)\Delta T \tag{7}$$

$$v(i+1) = v(i) + a(i)\Delta T \tag{8}$$

V. Distributed Model Predictive Controller Formulation

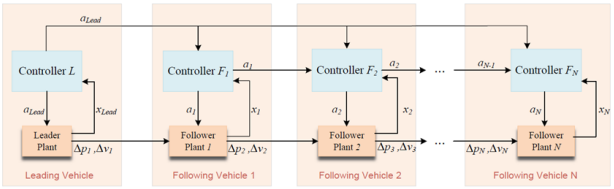


Fig. 4. DMPC Based ACC Platooning Control Structure

Distributed Model Predictive Control (DMPC) decomposes the overarching control problem into several sub-problems, each responsible for a subset of the control variables. In the context of multi-vehicle ACC for platooning showing in Fig 4, each vehicle independently optimizes its control actions. Coordination is achieved by exchanging information about predicted trajectories and control actions with neighbouring vehicles which is only the vehicle directly in front. This ensures a scalable and computationally efficient approach to manage large-scale traffic scenarios, maintaining safety and coordinated behaviour among all vehicles.

A. Cost Function

The control objective for the DMPC-based ACC within a platoon is represented by the following multi-objective function [4]:

$$J = \sum_{i=1}^{N} \Big[\alpha \cdot (d(t+i|t) - d_{\text{desired}})^2 + \gamma \cdot h(d(t+i|t)) \cdot (v(t+i|t) - v_{\text{pred},i})^2 + \delta \cdot \Delta u(t+i|t)^2\Big]$$

where:
d_{desired} is the targeted distance to the lead vehicle.
$h(d)$ is a threshold function that modulates based on the proximity to d_{desired}.
$v(t+i|t)$ and $v_{\text{pred},i}$ are the predicted speeds of the current and leading vehicle, respectively, at time $t+i$.
$d(t+i|t)$ denotes the predicted distance to the lead vehicle at that time.
$\Delta u(t+i|t)$ represents the change in control action.
α, γ, and δ are weight factors influencing distance maintenance, speed matching, and control smoothness, respectively.

B. Desired Distance Model

In various literature, the concept of the desired distance, often referred to as the safe distance, between two vehicles has been defined in several ways. One of the prevalent models for determining the desired distance is the constant time headway policy, which is described as:

$$d_{desired} = \tau_h v(t) + d_o \tag{9}$$

where τ_h is the nominal time headway, and s_o is the standstill distance. Furthermore, this work incorporates an additional model for the desired distance as:

$$d_{desired} = \frac{v^2}{2\mu g} + d_{buffer} \tag{10}$$

where the first term represents the minimum stopping distance necessitated by road friction (μ), and the second term is the buffer distance allocated for the following vehicle.

C. Threshold Function Model

The threshold function $h(d)$ modulates the importance of speed matching in the control algorithm based on the vehicle's distance d relative to the desired distance d_{desired}. It is defined mathematically as an exponential decay function:

$$h(d) = \alpha \cdot \exp\left(-\kappa \cdot \frac{|d - d_{\text{desired}}|}{d_{\text{desired}}}\right) \tag{11}$$

Here, α is a scaling factor that adjusts the overall impact of the function, and κ is the steepness factor, which dictates how quickly the function's value decays as the distance d diverges from d_{desired}. A higher value of κ results in a steeper decay, meaning that as the vehicle's distance deviates more from d_{desired}, the less impact the speed matching will have in the objective function.

D. Constraints

The constraints in the DMPC formulation are designed to ensure the physical limitations of the vehicle and the safety regulations of the road are adhered to. They include:

$$0 \leq T_d \leq 0.9 \cdot \mu \cdot m \cdot g \tag{12}$$

The maximum drive torques are limited by 90% of the product of the coefficient of road friction (μ), the mass of the vehicle (m), and the acceleration due to gravity (g)

$$s_{\text{safe}} \leq \Delta s = s_{\text{lead}} - s \tag{13}$$

where Δs is the distance between the lead vehicle and the ego vehicle, and s_{safe} is the safe distance that should not be undercut.

$$T_b \leq 0.9 \cdot \mu \cdot m \cdot g \tag{14}$$

The braking torque T_b (a negative value) is limited by the frictional capabilities of the road surface, ensuring safe deceleration.

$$0 \leq v \leq v_{\text{speedlimit}} \tag{15}$$

The velocity of the vehicle should not exceed the speed limit of the road ($v_{\text{speedlimit}}$).

VI. Centralized Model Predictive Control Formulation

Centralized Model Predictive Control (CMPC) provides a global optimization approach to multi-vehicle Adaptive Cruise Control (ACC) scenarios. Unlike Distributed Model Predictive Control (DMPC) that offers local optimization, CMPC leverages information from all vehicles to ensure a comprehensive perspective of the traffic dynamics, aiming for a global optimal solution.

Mathematically, the CMPC optimization can be defined as stacking the state and control variables of all vehicles in the scenario:

$$\mathbf{x} = [x_1^T, x_2^T, \ldots, x_N^T]^T$$
$$\mathbf{u} = [u_1^T, u_2^T, \ldots, u_N^T]^T$$

Where: - x_i represents the state variables of the ith vehicle. - u_i denotes the control variables of the ith vehicle. - N is the total number of vehicles.

The objective function for CMPC, based on the individual objective functions of the vehicles, is given by:

$$\begin{aligned} J_{\text{CMPC}} = \sum_{j=1}^{N} \sum_{i=1}^{P} [&\alpha \cdot (d_j(t+i|t) - d_{\text{desired}})^2 + \\ &\gamma \cdot h(d_j(t+i|t)) \cdot (v_j(t+i|t) - v_{\text{pred},j,i})^2 + \\ &\delta \cdot u_j(t+i|t)^2] \end{aligned}$$

Where N is the number of vehicles. P is the prediction horizon.

This unified objective function J_{CMPC} seeks to minimize the deviations of each vehicle's state from its desired state and the control efforts across the entire fleet of vehicles, emphasizing the centralized optimization approach of CMPC. The primary challenge of CMPC is the complexity involved in solving the optimization problem with the consolidated state and input variables, aiming for a consistent control trajectory across the fleet. The inherent nature of CMPC implies significant computational demands, particularly as the number of vehicles and the prediction horizon increase.

VII. Computational Cost Analysis of DMPC and Centralized MPC

In this section, we analyze and compare the computational complexities of Distributed Model Predictive Control (DMPC) and Centralized Model Predictive Control (CMPC) from both theoretical and mathematical perspectives.

A. Algorithm Structure

DMPC decentralizes the control problem, allowing each agent in the network to compute its control actions independently based on local information and coordination with neighboring agents. In contrast, CMPC centralizes the control computation, requiring a global model and comprehensive information about all agents.

B. Time Complexity

The time complexity of DMPC and CMPC can be characterized as follows:

- DMPC's time complexity is influenced by both the number of agents in the network and the complexity of computation per agent. It typically exhibits a time complexity of $O(n \cdot m^3)$, where n is the number of agents and m is the complexity of computation for each agent.
- CMPC, on the other hand, often has a time complexity of $O(p^3)$, where p is the total number of variables in the centralized optimization problem. This complexity arises from the need for global optimization over all agents, involving large-scale matrix operations.

C. Space Complexity

The space complexity for storing and processing data in both approaches is also crucial:

- In DMPC, the space complexity is generally $O(n \cdot m^2)$, reflecting the storage requirements for local models and information at each agent.
- For CMPC, the space complexity can be significantly higher, often $O(p^2)$, due to the need to store and process information about the entire network of agents.

D. Conclusion

DMPC offers scalability and reduced computational load per agent but with an overall complexity that grows with the number of agents. CMPC provides global optimization but at the expense of higher computational and storage requirements, especially in large networks. The choice between DMPC and CMPC depends on the specific requirements of the control system, including the number of agents, system coupling degree, and available computational resources.

VIII. RESULTS AND DISCUSSION

To assess the effectiveness of the proposed DMPC and CMPC controllers, a MATLAB simulation is employed to design and construct both the platoon model and controllers. The simulation involves a homogeneous platoon comprising one lead vehicle and three followers. The platoon has a desired spacing between each vehicle as 25 m, representing the distance from the preceding vehicle's rear end to the following vehicle's front end. The simulation considers constraints on torque, with a maximum limit of 400 Nm and a minimum limit of -400 Nm, applied to each vehicle in the platoon.

To conduct the simulation, the initial spacing between two consecutive vehicles is considered as 30 m. The leading vehicle uses an Urban Dynamometer Driving Schedule (UDDS) driving cycle is being used as the leading vehicle's velocity profile. The sampling time of simulation is 0.1 sec, and the prediction horizon of both CMPC and DMPC is considered to be 3. To depict the efficacy of CMPC and DMPC, the root mean distance between the leading and first following vehicle is considered.

CMPC and DMPC controllers are tuned in the simulation process by adjusting the weights assigned to distance, speed, and control input errors. The increment of the parameter contributes to a reduction in distance error, resulting in a smoother trajectory and decrement of the value for less aggressive control action. Fig 5 and Fig 6 illustrate the variation of the root mean distance between the leading and first following vehicles for different values of α. It is inferred from these figures that higher values of lead to the root mean distance approaching the desired spacing, and there is negligible distinction in the performance of CMPC and DMPC controllers. Following tuning, the values of α=100 and δ=0.001 are selected for both CMPC and DMPC, yielding reduced spacing error and a less aggressive control input.

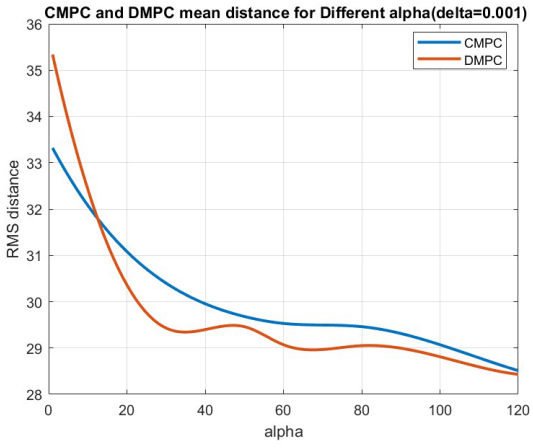


Fig. 5. Varition of RMS distance with α, for δ=0.001

By considering δ=0.001 and δ=0.01, the performance of CMPC and DMPC are shown in Fig 5 and Fig 6, respectively. The simulation showed that CMPC has a smoother trajectory than DMPC, with a root mean distance of 28.43 m and 29.07 m, respectively. As shown in Fig 8 and Fig 10, The control input(i.e., torque) shows a non-aggressive behavior in most parts of the simulation. It is easy to observe from the Fig 7 and Fig 9 that the following vehicles formed a platoon with a smooth distance variation between each vehicle for both CMPC and DMPC controllers.

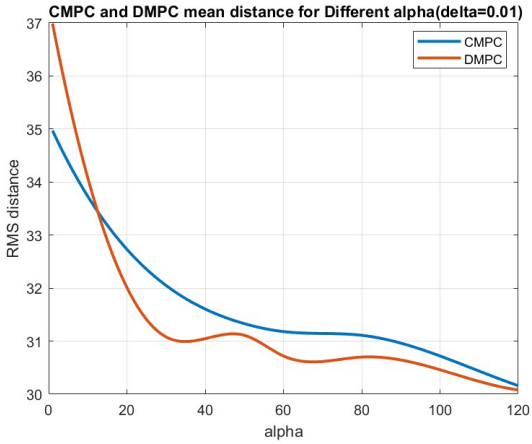


Fig. 6. Varition of RMS distance with α, for δ=0.01

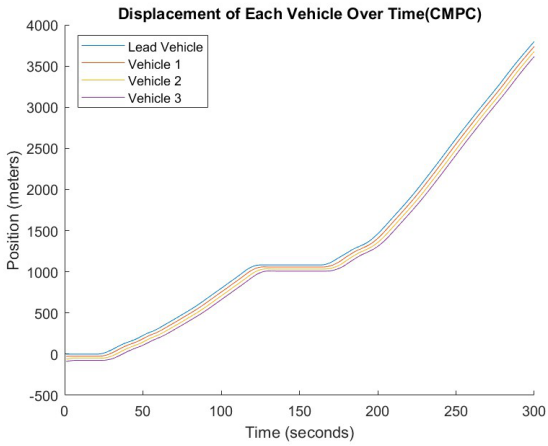


Fig. 7. Positions of vehicles in platoon using CMPC controller

In the control of multi-vehicle systems, one of the important factor to consider is computation time. The results indicate that the computation time for both CMPC and DMPC increases with the growing number of vehicles.

As discussed in Section 7, using CMPC to control multiple vehicles gives better performance based on the global cost function of the system. However, the communication and computational costs associated with implementing a centralized MPC may be high in multi-vehicle applications. Simulations demonstrate that optimal performance with less computation time can be achieved by employing DMPC. When the number of vehicles is two, the difference between the controllers is insignificant. However, for a system with more than two vehicles, the DMPC significantly reduces computational time

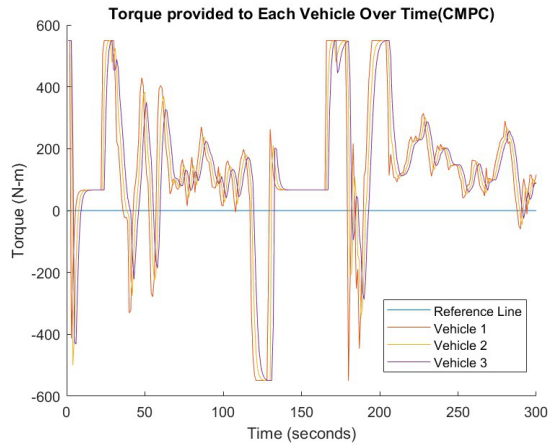


Fig. 8. Torque provided to vehicles in platoon using CMPC controller

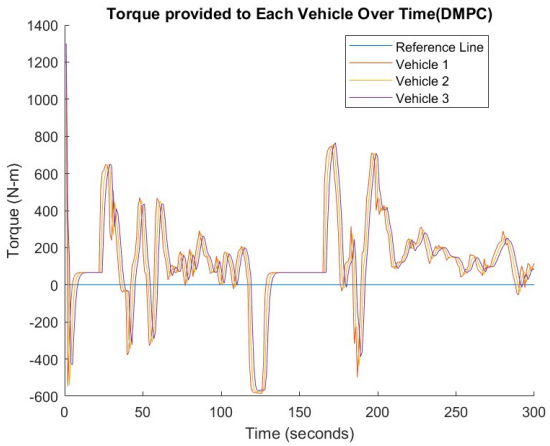


Fig. 10. Torque provided to vehicles in platoon using DMPC controller

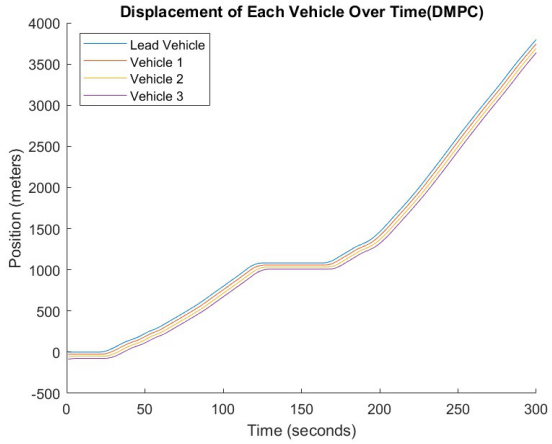


Fig. 9. Torque provided to vehicles in platoon using DMPC controller

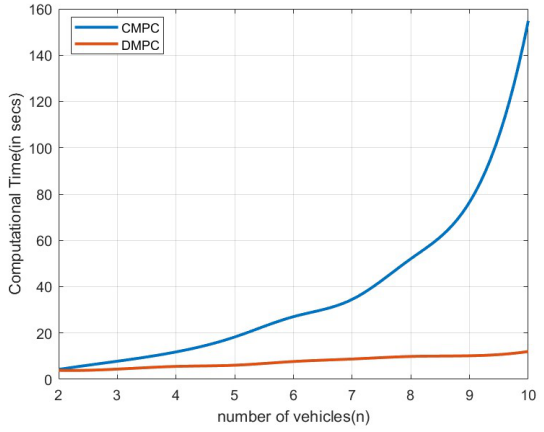


Fig. 11. Number of Vehicles Vs Computational Time

compared to the CMPC. Fig. 11 illustrates the computational time for CMPC and DMPC.

IX. CONCLUSION

This paper proposes two control schemes for vehicles in longitudinal traffic: CMPC and DMPC. CMPC treats all vehicles as a single optimization problem, while DMPC divides it into individual local problems for each car. This makes DMPC more flexible and scalable for larger traffic scenarios. Both CMPC and DMPC effectively managed to maintain space between vehicles, but CMPC offered a slightly smoother trajectory at the expense of higher computational cost, especially for larger platoon formations. As CMPC's computation time grew approximately quadratic with the number of vehicles, DMPC scaled linearly, making it more suitable for larger platoons. Therefore, DMPC provides a valuable trade-off between performance and computational cost, making it an attractive option for controlling large groups of vehicles.

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